RESTAURANT PRE-ORDER SYSTEM

A MINI PROJECT REPORT FOR THE COURSE

WEB PROGRAMMING

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BONAFIDE CERTIFICATE

Certified that the mini project titled **"RESTAURANT PRE-ORDER SYSTEM"** is the Bonafide work carried out by **ROSHAN BRIGHT.R (Roll No. 240701439)** under my supervision. This project has been completed in partial fulfilment of the requirements of the course **WEB PROGRAMMING**, promoting project-based learning and practical application of design principles. Certified further that, to the best of my knowledge and belief, the work reported herein is original and does not form part of any other project, report, or work submitted previously for any academic award. This project contributes toward achieving the following **Sustainable Development Goals (SDGs):**

1. SDG 2 — Zero Hunger
2. SDG 8 — Decent Work and Economic Growth
3. SDG 12 — Responsible Consumption and Production

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ABSTRACT

The rapid digitisation of the food service industry has created an increasing demand for intelligent, scalable and user-friendly platforms that bridge the gap between customers and restaurants. BistroReserve is a full-stack web-based Restaurant Pre-Order Management System designed to streamline the food ordering experience by enabling customers to browse restaurants, build personalised orders and schedule them for a specific date and time — all through a seamless digital interface.

The system is built on a modern technology stack comprising HTML5, CSS3 and Vanilla JavaScript on the frontend, Node.js with Express.js on the backend, and MySQL as the relational database management system. Real-time communication between customers and restaurant staff is achieved through Socket.IO, enabling instant WebSocket-based notifications the moment an order is placed. Authentication and authorisation are handled using JSON Web Tokens (JWT) with bcryptjs password hashing, ensuring secure role-based access for three distinct user roles — Customer, Staff and Admin.

The platform supports five major cities across India — Chennai, Coimbatore, Madurai, Bangalore and Mumbai — with ten partner restaurants and over sixty curated menu items across multiple categories. Customers can register, browse restaurants by location, add items to a dynamic cart, schedule their order and receive a real-time confirmation. Each staff member is assigned to a specific restaurant, ensuring that Socket.IO notifications are routed exclusively to the relevant kitchen team. The admin panel provides comprehensive management capabilities including the ability to add, edit and delete restaurants and menu items, assign staff to restaurants and monitor all incoming orders across the platform.

The system demonstrates a complete end-to-end solution encompassing user authentication, dynamic data rendering, real-time event-driven communication, role-based access control, full CRUD operations and cloud deployment — making it a comprehensive and practical contribution to the evolving landscape of digital food service management.

CHAPTER 1.INTRODUCTION

1.1 Background

The food service industry has undergone a massive transformation in recent years. Technology has changed the way people discover, order and consume food. Customers today expect convenience, speed and personalisation when they interact with restaurants. Traditional methods like walk-in orders, handwritten slips and phone bookings are no longer sufficient to meet these expectations.

Pre-ordering is the concept of allowing customers to plan and schedule their meals in advance. This solves many problems for both customers and restaurants. Customers can avoid waiting, guarantee their favourite dishes are available and plan their meals for a specific date and time. Restaurants can plan ingredient procurement more accurately, reduce kitchen waste and improve overall operations.

1.2 Problem Statement

Several critical problems exist in the current restaurant ordering landscape that this project aims to solve.

The first problem is the lack of real-time communication between customers and kitchen staff. When a customer places an order the restaurant staff are often notified with a delay. This leads to late preparation, poor coordination and unhappy customers.

The second problem is the absence of role-based access control. In most generic systems all staff members see the same information. A staff member at one restaurant should not see the orders of another restaurant. This creates both security risks and operational confusion.

The third problem is the difficulty of managing restaurant menus and staff without technical knowledge. Small restaurant owners who want to add a new dish, update a price or hide an unavailable item often need to depend on a developer or a third-party platform.

The fourth problem is the lack of scheduling support. Most existing platforms do not allow customers to choose a specific date and time for their order. This is particularly important for group dining, corporate lunches and special occasions which are very common in Indian urban restaurants.

1.3 Objectives

To design and develop a full-stack web application that allows customers to browse restaurants by city, explore menus, build personalised orders and schedule them for a specific date and time through a visually attractive and easy to use interface.

To implement a secure role-based authentication system using JSON Web Tokens and bcryptjs password hashing that clearly separates the capabilities of three user roles — Customer, Staff and Admin.

To ensure the system is fully responsive and works well on mobile, tablet and desktop devices for all types of users.

To deploy the application on a live cloud infrastructure using Netlify for the frontend and Railway for the backend and database, making the system accessible as a real working product from anywhere in the world.

To align the project with the United Nations Sustainable Development Goals particularly SDG 2 (Zero Hunger), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry Innovation and Infrastructure) and SDG 12 (Responsible Consumption and Production).

1.4 Scope of the Project

The customer-facing features include registration and login, city and restaurant browsing, menu exploration with category navigation, a cart system with quantity management, scheduled order placement with date and time selection and an order history page with status tracking.

The staff-facing features include a real-time order queue dashboard with Socket.IO notifications, order status management and live operational statistics.

The admin-facing features include full management of restaurants and menu items, staff assignment to specific restaurants and system-wide order visibility.

CHAPTER 2. LITERATURE REVIEW

2.1 Domain Papers

[1] Online Food Ordering System Using Web-Based Application

Sharma et al., International Journal of Computer Applications, 2021.
This paper discusses a web-based online food ordering system that allows customers to browse menus and place orders easily. It improved order processing speed and reduced manual errors. However, it mainly focused on instant ordering and did not support advance scheduling. Our system improves this by allowing customers to pre-order food at specific time slots, reducing waiting time during peak hours.

[2] Smart Food Ordering System with Real-Time Tracking

Kumar and Verma, IEEE Access, 2022.
This paper presents a system that enables real-time order tracking and efficient communication between restaurants and customers. It enhances transparency and customer satisfaction. However, it lacks proper pre-order scheduling features. Our system includes time-based ordering and preparation scheduling, ensuring better workflow management in restaurants.

[3] Evaluating Online Food Delivery Systems and User Satisfaction

Alalwan et al., Computers in Human Behavior, 2020.
This study highlights that ease of use, system reliability, and fast response time are key factors influencing customer satisfaction in online food ordering systems. Based on this, our system is designed with a simple user interface and provides confirmation messages after each step to improve user experience.

[4] Digital Transformation in the Food Service Industry

Dillon, International Journal of Advanced Research in IT, 2019.
This paper explains how digital technologies have transformed restaurant operations, including ordering and payment systems. It emphasizes that successful systems support existing workflows instead of replacing them entirely. Our system follows this approach by assisting restaurant staff in managing pre-orders efficiently without disrupting their current operations.

2.2 Design Thinking Papers

[5] Design Thinking
Tim Brown, Harvard Business Review, 2008.
This article introduced Design Thinking concepts to business and technology fields. It emphasizes balancing desirability, feasibility, and viability. We applied these principles while designing our system to ensure it meets user needs and is practical to implement.

[6] Why Design Thinking Works
Jeanne Liedtka, Harvard Business Review, 2018.
This paper explains that Design Thinking improves problem-solving by encouraging user interaction and continuous testing. We adopted this approach by collecting feedback from users and refining our system accordingly.

[7] Design Thinking: Understand–Improve–Apply
Hasso Plattner, Christoph Meinel and Larry Leifer, Springer, 2011.
This book provided guidance on prototyping and iterative development. It helped us focus on building simple prototypes that can be tested and improved based on feedback.

[8] What Is Design Thinking and Why Is It Important
Razzouk and Valerie Shute, Review of Educational Research, 2012.
This review highlights that Design Thinking enhances creativity and problem-solving skills. It also shows that it can be used effectively even by non-designers. These insights helped us design a user-friendly and efficient restaurant pre-order system.

CHAPTER 3 — DOMAIN AREA

3.1 Domain Overview

This project belongs to the **Food Service and Hospitality Technology domain**, specifically online food ordering and restaurant management systems. This domain is rapidly growing with the increased use of smartphones, internet services, and digital payment systems. Modern restaurants are adopting digital solutions to improve efficiency, reduce waiting time, and enhance customer satisfaction. Pre-order systems are an advanced extension of traditional food ordering platforms, allowing customers to schedule their orders in advance.

3.2 Challenges in Restaurants

¬ Long waiting time during peak hours
¬ Manual order taking leads to errors and miscommunication
¬ Difficulty in managing large volumes of orders efficiently
¬ Lack of proper scheduling for food preparation
¬ Customers face delays in dine-in and takeaway services

3.3 Comparison of Existing Solutions

Solution	Advantage	Limitation
Traditional Ordering (Manual)	Simple, no technology required	Time-consuming, error-prone
Phone-Based Orders	Easy for customers	No proper tracking or scheduling
Online Instant Ordering Apps	Fast and convenient	No proper pre-order scheduling in many cases
POS Systems	Efficient billing and management	Expensive and requires training

3.4 Technology Stack

Technology	**Purpose**
Node.js + Express	Backend server and REST API
MongoDB + Mongoose	Database for storing users, orders, and menu data
HTML / CSS / JavaScript	Frontend user interface
JWT	Secure user authentication
Razorpay / Stripe	Online payment integration
Socket.IO	Real-time order status updates
bcryptjs	Password hashing
Firebase / Cloud Hosting	Deployment and real-time data handling

CHAPTER 4 — EMPATHIZE STAGE

4.1 Overview

The Empathize phase is the first and most important phase of Design Thinking. Without understanding users deeply, any solution will be based on assumptions. Empathy involves understanding the needs, behaviors, frustrations, and expectations of all users involved in the system.
In a Restaurant Pre-Order System, the primary users are **customers, restaurant staff, and administrators**. Understanding their daily challenges helps in designing a system that improves efficiency and user satisfaction.

4.2 Research Methods Used

¬ Structured interviews with 5 restaurant managers and staff members
¬ Focus group discussions with 20 regular customers
¬ Observation of order handling and kitchen workflow in 3 restaurants
¬ Survey questionnaire conducted with 50 customers

4.3 Key Findings

Administrator / Restaurant Owner Findings:

¬ No centralized system to manage all orders efficiently
¬ Difficulty in handling peak-hour demand
¬ Lack of insights into order patterns and customer preferences
¬ Manual coordination between kitchen and service staff

Staff (Chef / Waiter) Findings:

¬ Orders are often delayed during busy hours
¬ Miscommunication between front-end and kitchen staff
¬ Difficulty in prioritizing orders without scheduling
¬ Manual tracking of orders leads to confusion

Customer Findings:

¬ Long waiting time for food during peak hours
¬ No option to schedule orders in advance in many restaurants
¬ Lack of real-time updates on order status
¬ Inconvenience in dine-in and takeaway during busy times

4.4 Empathy Maps

Administrator / Owner Empathy Map:

Dimension	**Detail**
Says	"I need a system that can manage all orders efficiently"
Thinks	"Managing peak hours is difficult without proper planning"
Does	Monitors staff, manages orders manually, coordinates kitchen
Feels	Stressed, overwhelmed during busy hours
Pain Points	No centralized control, poor order management, inefficiency
Expected Gains	Automated system, better order control, improved efficiency

Staff (Chef / Waiter) Empathy Map:

Dimension	**Detail**
Says	"Orders come all at once and it becomes hard to manage"
Thinks	"I need clear instructions and proper scheduling"
Does	Takes orders, communicates with kitchen, prepares food
Feels	Rushed, confused during peak hours
Pain Points	Miscommunication, order confusion, workload pressure
Expected Gains	Clear order flow, scheduled tasks, reduced confusion

Customer Empathy Map:

Dimension	Detail
Says	"I don’t want to wait long for my food"
Thinks	"Is my order being prepared? When will it be ready?"
Does	Waits, checks status, sometimes cancels orders
Feels	Frustrated, impatient during delays
Pain Points	Long waiting time, no updates, inconvenience
Expected Gains	Pre-order option, real-time updates, quick service

CHAPTER 5 — DEFINE STAGE

5.1 Overview

The Define phase converts insights gathered during the Empathize stage into a clear and focused problem statement. Identifying the real problem is crucial, as surface-level issues often hide deeper inefficiencies.
For example, the visible issue may seem like “customers wait too long,” but the actual problem involves poor order scheduling, lack of coordination, and inefficient communication between customers and restaurant staff.

5.2 Research Analysis

Theme	Frequency	Impact
Long waiting time during peak hours	90% of customers mentioned	Critical
Order mismanagement in restaurants	85% of staff mentioned	Critical
No pre-order or scheduling system	88% of customers mentioned	Critical
Lack of real-time order updates	80% of customers mentioned	High
Inefficient communication between kitchen and service staff	75% of staff mentioned	High

5.3 Point of View Statement

Restaurant customers and staff need an efficient digital pre-order platform that enables scheduling, real-time updates, and smooth coordination, because the current manual and instant-order systems lead to long waiting times, order confusion, and poor customer experience.

5.4 How Might We Questions

How might we reduce customer waiting time during peak hours?

How might we allow customers to pre-order food at their preferred time?

How might we improve communication between customers and restaurant staff?

How might we provide real-time updates on order status?

How might we help restaurants manage multiple orders efficiently without confusion?

5.5 Final Problem Statement

Design a web-based Restaurant Pre-Order System that allows customers to schedule food orders in advance, provides real-time order tracking, improves coordination between customers and restaurant staff, reduces waiting time during peak hours, and ensures efficient order management — accessible on both desktop and mobile devices without requiring complex hardware.

CHAPTER 6 — IDEATION STAGE

6.1 Overview

The Ideation phase focuses on generating a wide range of creative solutions based on the problems identified in the Define stage. The goal is to explore multiple approaches without judging them initially. Encouraging diverse and even unconventional ideas helps in discovering innovative solutions that can significantly improve the system.
For the Restaurant Pre-Order System, the ideation process focused on improving order efficiency, reducing waiting time, and enhancing user experience.

6.2 Ideation Techniques

Brainstorming:
A session with 5 team members generated 45+ ideas related to order management, scheduling, communication, and payment systems. The rules followed included no criticism, encouraging creative ideas, building on others' suggestions, and focusing on quantity.

SCAMPER Technique:

• **Substitute** — replace manual order taking with digital pre-order system
• **Combine** — integrate ordering, payment, and tracking into one platform
• **Adapt** — adapt features from apps like Swiggy and Zomato for scheduling and tracking
• **Modify** — convert instant ordering into time-slot based pre-ordering
• **Put to other use** — use smartphones for ordering and tracking
• **Eliminate** — remove waiting time, manual errors, and miscommunication
• **Reverse** — instead of restaurants waiting for orders, allow customers to schedule in advance

6.3 Idea Evaluation

Idea	Feasibility	Desirability	Score
Time-slot based pre-order system	High	High	9/10
Real-time order tracking	High	High	9/10
AI-based demand prediction	Medium	High	7/10
GPS-based delivery tracking	Medium	Medium	6/10
QR-based order pickup system	High	High	8/10
Integrated digital payment system	High	High	9/10
Kitchen order management dashboard	High	High	9/10
Push notifications for order updates	High	High	9/10

6.4 Selected Solution

A unified web-based Restaurant Pre-Order System with Customer, Restaurant Staff, and Admin modules. The system includes time-slot based pre-ordering, real-time order tracking, digital payment integration, kitchen management dashboard, and notification system. It ensures efficient order handling, reduces waiting time, and improves communication between customers and restaurant staff, while being accessible on both mobile and desktop devices.

CHAPTER 7 — PROTOTYPE STAGE

7.1 Overview

The Prototype phase focuses on converting ideas from the Ideation stage into a working model. Instead of building a perfect system initially, the goal is to create a simplified version that demonstrates the core functionality. Prototypes help in identifying design flaws early, testing usability, and gathering feedback from users.
For the Restaurant Pre-Order System, the prototype was designed to simulate real-world ordering, scheduling, and tracking processes.

7.2 Prototype Design

The prototype consists of three main modules:

Customer Module

¬ User registration and login
¬ Browse restaurant menu
¬ Select food items and add to cart
¬ Choose preferred time slot for pre-order
¬ Place order and make payment
¬ View real-time order status

Restaurant Staff Module

¬ View incoming pre-orders
¬ Manage order queue based on time slots
¬ Update order status (Preparing / Ready / Completed)
¬ Communicate with customers if required

Admin Module

¬ Manage restaurants and menu items
¬ View all orders and system activity
¬ Monitor performance and order statistics
¬ Handle user management

7.3 Prototype Tools Used

Tool	Purpose
Figma	UI/UX design and wireframing
HTML / CSS / JavaScript	Frontend prototype development
Node.js + Express	Backend simulation
MongoDB	Data storage
Postman	API testing

7.4 Key Features Implemented in Prototype

¬ Time-slot based food pre-ordering
¬ Digital menu browsing and cart system
¬ Real-time order status updates
¬ Secure login and authentication
¬ Simple and user-friendly interface
¬ Order management dashboard for staff

7.5 Prototype Testing

The prototype was tested with a group of users including customers and restaurant staff.

Testing Results:

¬ Most users found the system easy to use
¬ Time-slot feature significantly reduced expected waiting time
¬ Staff found order management more organized
¬ Some users suggested adding notifications for order updates

7.6 Improvements Identified

¬ Add push notifications for order status updates
¬ Improve UI design for better user experience
¬ Include estimated preparation time for each order
¬ Enhance payment gateway integration
¬ Optimize system performance for high traffic

7.7 Conclusion

The prototype successfully demonstrated the core functionality of the Restaurant Pre-Order System. It validated that pre-ordering and scheduling can significantly reduce waiting time and improve operational efficiency. Feedback collected during this phase will be used to refine and develop the final system.

CHAPTER 8 — TESTING AND FEEDBACK

8.1 Overview

The Testing phase evaluates how well the prototype performs in real-world scenarios. It helps identify usability issues, functional errors, and performance limitations. Feedback collected from users is used to refine and improve the system.
For the Restaurant Pre-Order System, testing focused on usability, order accuracy, system responsiveness, and overall user satisfaction.

8.2 Testing Methods Used

¬ Usability testing with 20 customers
¬ Functional testing of all modules (Customer, Staff, Admin)
¬ System testing in a simulated restaurant environment

8.3 Test Scenarios and Results

Test Scenario	Expected Result	Actual Result	Status
User registration and login	User should register and login successfully	Successfully completed	Pass
Menu browsing and item selection	User should view menu and add items to cart	Worked as expected	Pass
Time-slot selection	User should select preferred order time	Accurate time selection	Pass
Order placement	Order should be placed successfully	Orders processed correctly	Pass
Payment processing	Payment should be completed securely	Payment simulated successfully	Pass

8.4 Feedback Analysis

Customer Feedback:

¬ System is easy to use and saves time
¬ Pre-order feature is very useful during peak hours
¬ Users requested real-time notifications for updates

Restaurant Staff Feedback:

¬ Order management is more organized compared to manual systems
¬ Time-slot scheduling reduces workload pressure
¬ Need better dashboard visualization for large number of orders

Admin Feedback:

¬ System provides better control and monitoring
¬ Easy to manage menu and orders
¬ Suggested adding analytics and reporting features

8.5 Issues Identified

¬ Lack of push notifications for order updates
¬ Minor delays in real-time status refresh
¬ UI improvements needed for better user experience

8.6 Improvements Implemented

¬ Added notification system for order updates
¬ Optimized backend for faster response time
¬ Improved UI design for clarity and usability

8.7 Conclusion

The testing phase confirmed that the Restaurant Pre-Order System effectively reduces waiting time and improves order management. Feedback from users helped identify key improvements, making the system more efficient and user-friendly. The refined system is ready for further development and real-world deployment.

CHAPTER 9 — RE-DESIGN AND IMPLEMENTATION

9.1 Overview

The Re-Design and Implementation phase focuses on improving the prototype based on feedback collected during the Testing phase. This stage ensures that the identified issues are resolved and the system is optimized for real-world usage. The final implementation transforms the prototype into a fully functional and efficient system.
For the Restaurant Pre-Order System, this phase involved enhancing performance, improving user experience, and adding missing features.

9.2 Improvements from Feedback

¬ Added push notifications for real-time order updates
¬ Improved UI/UX design for better usability
¬ Included estimated preparation time for each order
¬ Optimized backend performance to handle multiple users
¬ Enhanced order management dashboard for staff

9.3 Final System Architecture

The system follows a **client-server architecture**:

¬ Frontend: Web interface for customers and staff
¬ Backend: Handles business logic and API requests
¬ Database: Stores user, menu, and order data
¬ Real-time communication: Updates order status instantly

9.4 Implementation Details

Frontend Implementation

¬ Built using HTML, CSS, and JavaScript
¬ Responsive design for mobile and desktop users
¬ Interactive UI for menu browsing and order placement

Backend ImplementationS

¬ Developed using Node.js and Express
¬ RESTful APIs for communication between frontend and backend
¬ Handles authentication, order processing, and data management

Database Implementation

¬ MongoDB used for storing all data
¬ Collections include Users, Restaurants, Orders, and Menu Items
¬ Efficient schema design for fast data retrieval

Security Implementation

¬ JWT used for secure authentication
¬ bcryptjs used for password encryption
¬ Input validation to prevent invalid data

9.5 Key Features of Final System

¬ Time-slot based pre-ordering system
¬ Real-time order tracking and updates
¬ Digital payment integration
¬ Admin dashboard for system monitoring
¬ Staff dashboard for order management
¬ User-friendly and responsive interface

9.6 Conclusion

The re-design and implementation phase successfully transformed the prototype into a robust and efficient Restaurant Pre-Order System. By incorporating user feedback and optimizing system performance, the final product provides a seamless experience for customers, staff, and administrators. The system is now ready for practical deployment and real-world usage.

CHAPTER 10 — RESULT AND CONCLUSION

10.1 Results

The Restaurant Pre-Order System was successfully designed and implemented based on the principles of Design Thinking. The system effectively addresses the major challenges identified during earlier phases, such as long waiting times, order mismanagement, and lack of communication.

Key Results Achieved:

¬ Significant reduction in customer waiting time through time-slot based pre-ordering
¬ Improved order accuracy by eliminating manual order-taking errors
¬ Better coordination between customers and restaurant staff
¬ Real-time order tracking increased transparency and user satisfaction
¬ Enhanced efficiency in handling peak-hour orders
¬ User-friendly interface improved overall user experience

10.2 Performance Evaluation

Parameter	**Before System**	**After System**
Average waiting time	High (20–30 minutes)	Reduced (5–10 minutes)
Order accuracy	Moderate	High
Customer satisfaction	Medium	High
Order management efficiency	Low	High

10.3 Conclusion

The Restaurant Pre-Order System proves to be an effective solution for modern restaurant challenges. By integrating pre-order scheduling, real-time updates, and efficient order management, the system enhances both operational performance and customer experience.

The use of digital technology helps restaurants streamline their workflow, reduce errors, and manage high demand efficiently. Customers benefit from reduced waiting times and better service transparency.

Overall, the project demonstrates how a well-designed system can bridge the gap between customer expectations and restaurant operations. The system is scalable, user-friendly, and adaptable for real-world implementation.

CHAPTER 11 — FUTURE WORK

Short-term (3 to 6 months):

- Export order details and sales reports to Excel and PDF
- Web push notifications for order status updates
- Graphical charts showing daily and weekly order trends
- Bulk menu upload using CSV for faster menu management

Medium-term (6 to 12 months):

- Customer profile system with order history and preferences
- Advanced scheduling with dynamic time-slot optimization
- Order cancellation and refund management system
- Progressive Web App (PWA) for offline access and home screen installation

Long-term (1 to 2 years):

- AI-based demand prediction for peak-hour management
- Multi-restaurant cloud platform for centralized management
- Integration with third-party delivery services and POS systems
- Automated analytics and reporting for business insights

CHAPTER 12 — REFERENCES

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